POWER BI PRACTICAL ASSESSMENT

LumaCare Health & Wellness Services

Assessment	Individual Practical Assessment
Suggested Duration	2 Hours
Total Marks	100
Dataset	LumaCare_PowerBI_Assessment_Dataset.xlsx

Assessment Goal
Transform a deliberately imperfect business dataset into a clean, interactive Power BI dashboard and use it to make evidence-based management recommendations.

1. Case Study

LumaCare Health & Wellness Services operates branches in Colombo, Kandy, Galle and Jaffna. The company provides Wellness, Diagnostics, Dental and Physiotherapy services through multiple booking channels. Senior management believes revenue is growing, but needs a clearer view of profitability, branch performance, service performance and customer experience.

You have been appointed as a **Business Intelligence Analyst**. Using the supplied dataset, clean and prepare the data, create appropriate calculations, build an interactive executive dashboard, and present evidence-based recommendations to management.

Important: The dataset intentionally contains duplicates, missing values and inconsistent labels. Students are expected to identify and resolve these issues independently.

2. Part A - Data Preparation

20 marks

- Import the Transactions dataset into Power BI and open it in Power Query.
- Profile the dataset and identify the important data-quality problems.
- Remove duplicate transaction records appropriately.
- Handle missing Consultant and Satisfaction Score values using a justified method.
- Standardize inconsistent Branch and Service Category values.
- Confirm appropriate data types for all fields.
- Create transaction-level columns for **Revenue**, **Total Cost** and **Profit**.

3. Part B - DAX Measures

20 marks

- Create a measure for **Total Revenue**.
- Create a measure for **Total Cost**.
- Create a measure for **Total Profit**.
- Create a measure for **Profit Margin %**.
- Create a measure for **Total Transactions**.
- Create a measure for **Total Service Units**.
- Create a measure for **Average Satisfaction Score**.

4. Part C - Interactive Dashboard

30 marks

- Four KPI cards: Total Revenue, Total Profit, Profit Margin %, and Average Satisfaction Score.
- A monthly Revenue and Profit trend.
- Revenue by Branch.
- Profit Margin % by Service Category.
- Top 5 Service Packages by Revenue.
- Booking Channel performance.
- Customer Segment comparison.
- At least four useful slicers, including Date and Branch.

5. Part D - Business Analysis

20 marks - Use evidence from your dashboard when answering each question.

1. Which branch generates the highest revenue?

2. Does the highest-revenue branch also have the strongest profit margin?

3. Which service category contributes the most revenue?

4. Which service category appears financially inefficient when revenue and margin are considered together?

5. Which booking channel produces the strongest profitability?

6. Which booking channel has the highest average customer satisfaction?

7. Which month shows the strongest overall performance? Explain using more than one KPI.

8. Which three service packages should management prioritize and why?

9. Identify one branch/service combination that management should investigate.

10. Identify one insight that would be hidden if management considered revenue alone.

6. Part E - Management Recommendations

10 marks

Provide **three management recommendations**. Each recommendation must use the following structure:

Finding - What did you discover?
Evidence - Which KPI, comparison or visual supports the finding?
Recommended Action - What should LumaCare management do next?

Recommendations should be specific, actionable and directly supported by the dashboard.

7. Submission Requirements

- Power BI .pbix file.
- One PDF report containing a screenshot of the final dashboard.
- Screenshots of important Power Query cleaning steps.
- All seven DAX measures.
- Answers to all ten business-analysis questions.
- Three evidence-based management recommendations.
- Written interpretation - do not submit charts without explaining what they mean.

8. Marking Rubric

Assessment Area	Marks	Assessment Focus
Data Preparation	20	Cleaning accuracy, transformations, data types and calculated columr
DAX Measures	20	Correct measures, logic, aggregation and formatting
Dashboard	30	Visual selection, accuracy, layout, readability and interactivity
Business Analysis	20	Evidence, interpretation, comparison and business reasoning
Recommendations	10	Actionability and direct connection to dashboard evidence

Assessment Area	Marks	Assessment Focus
TOTAL	**100**	

Expected workflow: Raw Data -> Data Cleaning -> Calculations -> Dashboard -> Analysis -> Management Decision